FIITJEE

ALL INDIA TEST SERIES

FULL TEST – VIII

JEE (Advanced)-2019

PAPER – 2

Time Allotted: 3 Hours **Maximum Marks: 180**

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections**: **Section-A & Section-D.**

1. ***Section-A (01 – 06, 19 – 24, 37 – 42)*** *contains 18 multiple choice questions which have* ***one or more than one correct*** *answer. Each question carries* ***+4 marks*** *for all correct answer.*
   *Partial Marks : **+3** If all the four options are correct but ONLY three options are chosen.*
   *Partial Marks : **+2** If three or more options are correct but ONLY two options are chosen, both of which are correct options.*
   *Partial Marks : **+1** If two or more options are correct but ONLY one option is chosen and it is a correct option.*
   *Zero Marks : **0** If none of the options is chosen (i.e. the question is unanswered).*
   *Negative Marks : **–2** In all other cases.*

   Section-A (07 – 10, 25 – 28, 43 – 46) *contains 12 questions. Each question has* ***TWO (02)*** *matching lists:* ***LIST-I*** *and* ***LIST-II****.* ***FOUR*** *options are given representing matching of elements from* ***LIST-I*** *and* ***LIST-II****.* ***ONLY ONE*** *of these four options corresponds to a correct matching.*
   For each question, choose the option corresponding to the correct matching.
   Each question has ***only one correct*** *answer and carries* ***+3 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

2. ***Section-D (11 – 18, 29 – 36, 47 – 54)*** *contains 24 Numerical answer type questions with answer XXXXX.XX and each question carries* ***+3 marks*** *for correct answer. There is no negative marking.*

Physics

PART – I

SECTION – A
(One or More than one correct type)

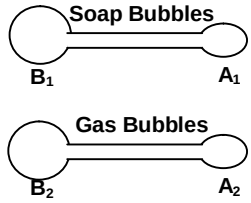
This section contains **6** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

1. Two pendulums of same amplitude but time period 3s and 7s start oscillating simultaneously from two opposite extreme positions. After how much time they will be in phase?

(A) $\frac{21}{8}$s (B) $\frac{21}{4}$s

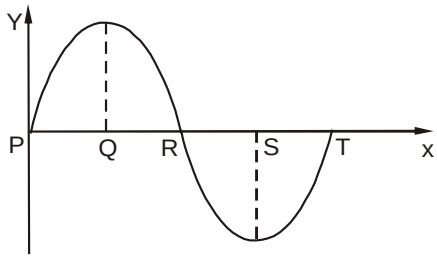
(C) $\frac{21}{2}$s (D) $\frac{63}{8}$s

2. A_1 and B_1 are two bubbles joined by a thin tube, similarly A_2 and B_2 are rubber balloons filled with air joined by another thin tube (Fig shows the situation) As time progress _____choose the correct statement (A_1 and A_2 are smaller in volume, then B_1 and B_2 respectively).



(A) A_1 will shrink : B_2 will shrink
(B) A_1 will shrink : A_2 will expand
(C) A_1 will expand : B_2 will shrink
(D) A_1 will expand : B_2 will expand

3. A sound wave is traveling along positive *x*-direction. Displacement (*y*) of particles from their mean positions at any time *t* is shown in the figure.



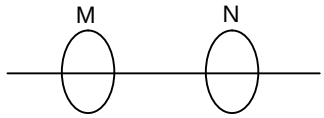
(A) Particle located at S has zero velocity.
(B) Particle located at T has its velocity in the negative direction.
(C) Change in pressure at S is zero
(D) Particles located near R are under compression.

Space for Rough work

4. A convex lens made of glass (μ_g =3/2) has focal length *f* in air. The image of an object placed in front of it is inverted real and magnified. Now the whole arrangement is immersed in water (μ_w=4/3) without changing the distance between object and lens. Then
(A) the new focal length will become 4*f*
(B) the new focal length will become *f* / 4
(C) new image will be virtual and magnified
(D) new image will be real, inverted and smaller in size

5. If the potential difference of Coolidge tube producing X-ray is increased, then choose the correct option(s).
(A) the interval between $\lambda_{k\alpha}$ and $\lambda_{k\beta}$ increases
(B) the interval between $\lambda_{k\alpha}$ and λ_0 increases
(C) the interval between $\lambda_{k\beta}$ and λ_0 increases
(D) λ_0 does not change
Here λ_0 is cutoff wavelength and $\lambda_{k\alpha}$ and λ_{kB} are wavelength of k_α and k_B characteristic X-rays.

6. Two identical circular coils M and N are arranged coaxially as shown in the figure. Separation between the coils is large as compared to their radii. The arrangement is viewed from left along the common axis. The sign convention adopted is that currents are taken to be positive when they appear to flow in clockwise direction. Then



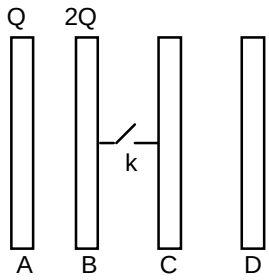
(A) if M carries a constant positive current and is moved towards N, a positive current is induced in N
(B) if M carries a constant positive current and N is moved towards M, a negative current is induced in N
(C) if a positive current in M is switched off, a positive current is momentarily induced in N
(D) if both coils carry positive currents, they will attract each other

Space for Rough work

(Matching List Type)

This section contains **FOUR** questions. Each question has **TWO** matching lists: **LIST - I** and **LIST - II**. **FOUR** options are given representing matching of elements from **LIST - I** and **LIST - II**. **ONLY ONE** of these four options corresponds to a correct matching.

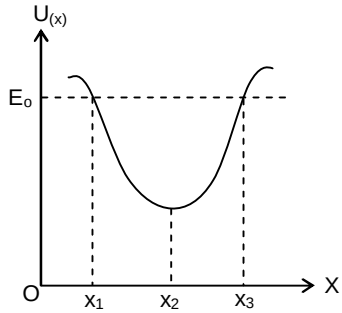
7. Read the following and write the correct pairs.
Initially, the two plates (A and B) are given charges as shown. Find the final charges on the plates after the key k is closed. All plates are conducting, parallel and of infinite length and breadth.



List-I		List-II	
(P)	Final charge on plate A	(1)	Q/2
(Q)	Final charge on plate B	(2)	Q
(R)	Final charge on plate C	(3)	0
(S)	Final charge on plate D	(4)	3Q/2

Codes:
(A) P–2, Q–1, R–4, S–3 (B) P–2, Q–4, R–1, S–3
(C) P–3, Q–1, R–4, S–2 (D) P–2, Q–3, R–2, S–4

8. The potential energy of a particle moving along x-axis is show in figure. Where E_o is the total mechanical energy of the particle.

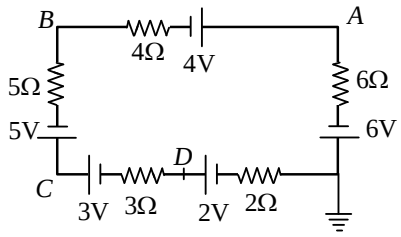


List-I		List-II	
(P)	Force on the particle is towards +ve x-axis if	(1)	$x_1 < x < x_2$
(Q)	Particle can't be found where	(2)	$x_2 < x < x_3$
(R)	Force is towards the equilibrium position if	(3)	$x < x_1, x > x_3$
(S)	Force is towards the –ve x-axis if	(4)	$x_2 > x$
		(5)	$x_1 < x < x_3$

Codes:
(A) P–2, Q–1, R–3,4, S–2,3 (B) P–1,4, Q–3, R–1,2 4,5, S–2
(C) P–2,3, Q–3, R–1,4, S–2,3 (D) P–2,3,4, Q–1, R–2, S–2,4

Space for Rough work

9. Five batteries whose emf and internal resistance are as shown in figure. Match the following



List–I		List–II	
(P)	Potential of point *A*	(1)	– 5V
(Q)	Potential of point *B*	(2)	1 V
(R)	Potential of point *C*	(3)	– 3V
(S)	Potential of point *D*	(4)	2.5V
		(5)	1.5 V

Codes:

(A) P–3, Q–4, R–3, S–1 (B) P–2, Q–4, R–1, S–3

(C) P–3, Q–1, R–4, S–2 (D) P–2, Q–3, R–2, S–4

10. The speed (v) of a particle of mass 1 kg moving along a straight line, when it is at a distance (x) from a fixed point on the line, is given by, $v^2 = 144 - 9x^2$. Match the following:

List–I		List–II	
(P)	Motion is simple harmonic with time period	(1)	$2\pi/3$ units
(Q)	Max. displacement from the fixed point is	(2)	27 units
(R)	Magnitude of acceleration at a distance 3 units from the fixed point is	(3)	4 units
(S)	Potential energy at a distance x = 2 unit from fixed point is	(4)	2 units
		(5)	18 units

Codes:

(A) P–1, Q–4, R–2, S–3 (B) P–1, Q–3, R–5, S–2,5

(C) P–3, Q–1, R–2, S–5 (D) P–1, Q–3, R–2, S–5

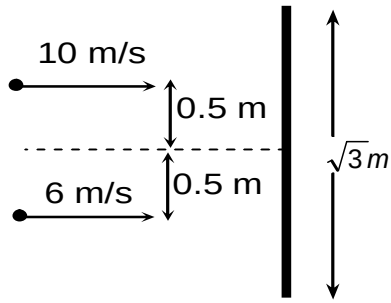
Space for Rough work

SECTION – D
(Numerical Answer Type)

This section contains **EIGHT** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. xxxxx.xx).

11. Gravitational acceleration on the surface of a planet is $\frac{\sqrt{6}}{11}g$, where g is the gravitational acceleration on the surface of the earth. The average mass density of the planet is $\frac{2}{3}$ times that of the earth. If the escape speed on the surface of the earth is taken to be 11 km s^{-1}, the escape speed on the surface of the planet in kms^{-1} will be

12. A thin uniform bar lies on a frictionless surface and is free to move in any way on the surface. Its mass is 0.16 kg and length $\sqrt{3}$ metres. Two particles each of mass 0.08 kg are moving on the same surface and towards the bar in a direction perpendicular to the bar, one with a velocity 10 m/s and the other with 6 m/s as shown in Figure. The first particle strikes the bar at *A* and the other at the point B. The points A and B are at distances 0.5 m from centre of the bar. The particles strike the bar at the same instant of time and stick to the bar on collision. Calculate the loss of kinetic energy of the system on the above collision process.

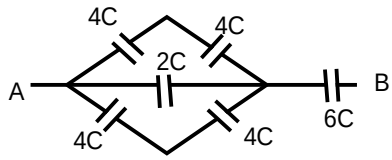


13. A light wooden rod fixed at one end is kept horizontal. A load of 0.4 kg tied to the free end of the rod causes that end to be depressed by 2.8 cm. If this load is set into up and down vibrations, with what frequency will it oscillate?

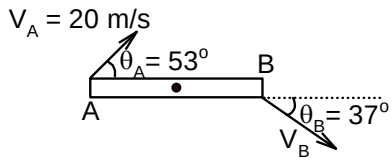
Space for Rough work

14. A moving car encounters air resistance which is proportional to the square of the speed of the car. Find the value of $\frac{2 \times \text{Power required at 40 kmph}}{\text{Power required at 80 kmph}}$

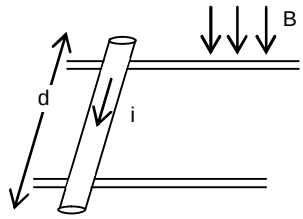
15. The equivalent capacitance between points A and B of the circuit will be (given: C = 1.3)



16. The two ends A and B of a uniform rod of length ℓ = 1m are moving with velocities V_A and V_B as shown. The velocity V_B (m/s) is



17. A cylindrical uniform rod of mass 0.72 kg and radius 6 cm rests on two parallel rails, that are d = 50 cm apart. The rod caries a current I = 48A (In the direction shown) and rolls along the rails without slipping. If it starts from rest, uniform magnetic field of magnitudes 0.25 T is directed perpendicular to the rod and the rail, then find the friction force(In N) between rod and rails



18. A 10 m wide spacecraft moves through the interstellar space at a speed 3×10^7 m/s. A magnetic field $B = 3 \times 10^{-10}$ T exists in the space in a direction perpendicular to the plane of motion. Treating the spacecraft as a conductor, calculate the emf induced across its width. (in volt)

Space for Rough work

Chemistry

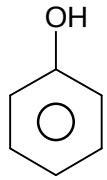
PART – II

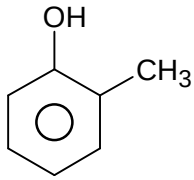
SECTION – A
(One or More than one correct type)

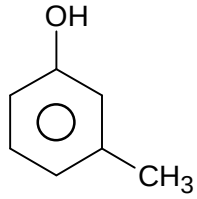
This section contains **6** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

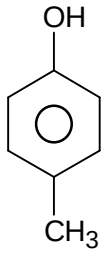
19. Which of the following substance(s) is/are more soluble than $CaSO_4$ in water?
(A) $BeSO_4$ (B) $MgSO_4$
(C) $SrSO_4$ (D) $BaSO_4$

20. Which of the following compound(s) form(s) tribromo derivative(s) as the major product when treated with bromine water?

(A) 

(B) 

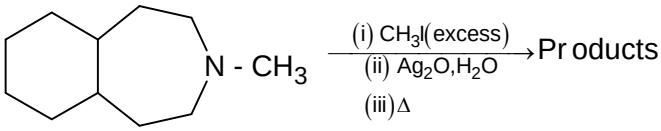
(C) 

(D) 

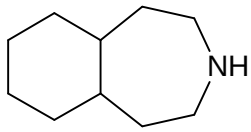
21. Which of the following is/are the anhydride(s) of boric acid(H_3BO_3)?
(A) B_2O_3 (B) HBO_2
(C) $H_2B_2O_3$ (D) $H_2B_4O_7$

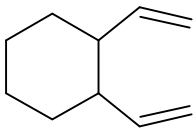
Space for Rough work

22. The molecular formula of which of the following is/are $C_4H_4O_4$?
(A) A cis-dibasic acid
(B) An α, β unsaturated dibasic acid which on heating gives a monobasic acid
(C) A cycic ester of dibasic acid
(D) An keto ester

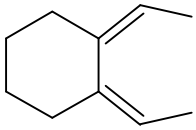
23. 

Which of the following product(s) is/are formed in the above reaction?

(A) 

(B) 

(C) $(CH_3)_3N$

(D) 

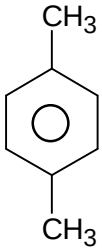
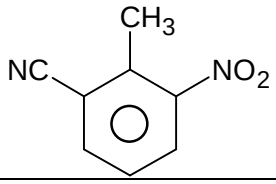
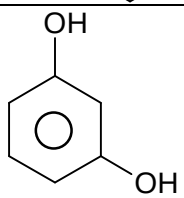
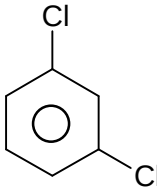
24. Which of the following statement(s) about aminoacids is/are correct?
(A) They are all high melting solids
(B) They are constituents of all proteins
(C) Glycine is the only naturally occurring amino acid which is optically inactive
(D) Most naturally occurring amino acids have D-configuration

Space for Rough work

(Matching List Type)

This section contains **FOUR** questions. Each question has **TWO** matching lists: **LIST - I** and **LIST - II**. **FOUR** options are given representing matching of elements from **LIST - I** and **LIST - II**. **ONLY ONE** of these four options corresponds to a correct matching.

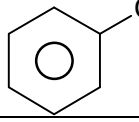
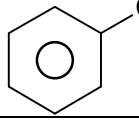
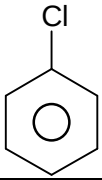
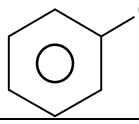
25. Match the compounds mentioned in List-I with their properties mentioned in List-II.

List – I		List – II	
(P)		(1)	Forms one monosubstituted product in electrophilic substitution reactions.
(Q)		(2)	Forms more than one monosubstituted product in electrophilic substitution reactions
(R)		(3)	More reactive than benzene towards EAS reaction
(S)		(4)	Less reactive than benzene towards EAS reaction
		(5)	The substituents exert –I as well as +R effect

(A) P $\rightarrow$ 1, 3; Q $\rightarrow$ 1, 4; R $\rightarrow$ 2, 4, 5; S $\rightarrow$ 2, 4, 5
(B) P $\rightarrow$ 3; Q $\rightarrow$ 3, 5; R $\rightarrow$ 1, 2 ; S $\rightarrow$ 1, 3
(C) P $\rightarrow$ 2; Q $\rightarrow$ 4; R $\rightarrow$ 1, 3; S $\rightarrow$ 5
(D) P $\rightarrow$ 4, 5; Q $\rightarrow$ 2; R $\rightarrow$ 5; S $\rightarrow$ 1, 5

Space for Rough work

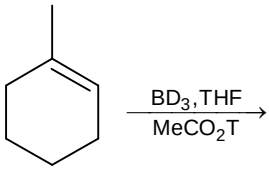
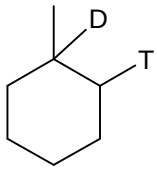
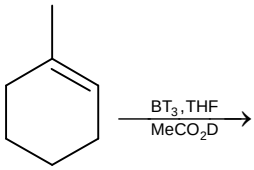
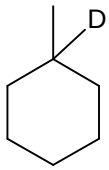
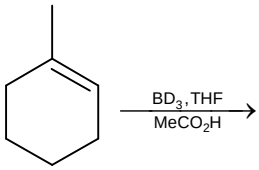
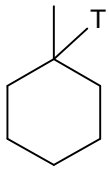
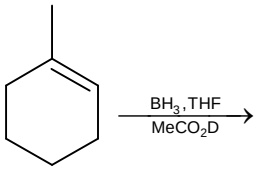
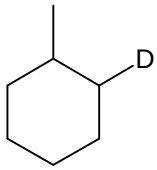
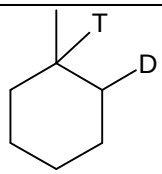
26. Match the compounds mentioned in List – I with their properties mentioned in List – II.

List – I		List – II	
(P)		(1)	More reactive than toluene towards common electrophilic bromination reaction in water
(Q)		(2)	Forms carbocation which is stabilized through resonance
(R)		(3)	Turns blue litmus to red
(S)		(4)	The substituted atom or group shows +R effect
		(5)	The substituted atom or group shows –I effect

(A) P $\rightarrow$ 2, 3; Q $\rightarrow$ 5; R $\rightarrow$ 4; S $\rightarrow$ 1
(B) P $\rightarrow$ 4; Q $\rightarrow$ 3; R $\rightarrow$ 1, 2; S $\rightarrow$ 1, 3
(C) P $\rightarrow$ 1, 3, 4, 5; Q $\rightarrow$ 1, 4, 5; R $\rightarrow$ 4, 5; S $\rightarrow$ 2, 5
(D) P $\rightarrow$ 3; Q $\rightarrow$ 1, 4; R $\rightarrow$ 2, 3, 5; S $\rightarrow$ 2, 4, 5

Space for Rough work

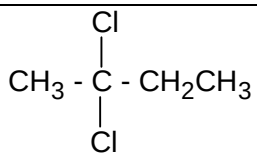
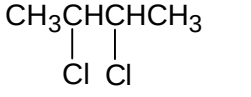
27. Match the following:

List – I		List – II	
(P)		(1)	
(Q)		(2)	
(R)		(3)	
(S)		(4)	
		(5)	

(A) P $\rightarrow$ 3, Q $\rightarrow$ 1, R $\rightarrow$ 4, S $\rightarrow$ 5
(B) P $\rightarrow$ 1; Q $\rightarrow$ 5; R $\rightarrow$ 2; S $\rightarrow$ 4
(C) P $\rightarrow$ 5, Q $\rightarrow$ 4, R $\rightarrow$ 3, S $\rightarrow$ 2
(D) P $\rightarrow$ 4, Q $\rightarrow$ 2, R $\rightarrow$ 5, S $\rightarrow$ 1

Space for Rough work

28. Match the substances mentioned in List-I with their properties mentioned in List-II.

List – I		List – II	
(P)	$CH_3CH_2CH_2CHCl_2$	(1)	Forms only terminal alkyne on dehydrohalogenation reaction with $NaNH_2$ followed by H_2O
(Q)		(2)	Forms only internal alkyne on dehydrohalogenation reaction with $NaNH_2$ followed by H_2O
(R)		(3)	Forms both internal and terminal alkyne on dehydrohalogenation reaction with $NaNH_2$ followed by H_2O
(S)	$ClCH_2CH_2CH_2CH_2Cl$	(4)	Forms conjugated diene on dehydrohalogenation reaction

(A) P $\rightarrow$ 2, 3; Q $\rightarrow$ 1, 2; R $\rightarrow$ 4; S $\rightarrow$ 1
(B) P $\rightarrow$ 3; Q $\rightarrow$ 4; R $\rightarrow$ 1, 2; S $\rightarrow$ 4
(C) P $\rightarrow$ 4, Q $\rightarrow$ 3, R $\rightarrow$ 3, S $\rightarrow$ 2
(D) P $\rightarrow$ 1; Q $\rightarrow$ 3; R $\rightarrow$ 2, 4; S $\rightarrow$ 4

SECTION – D
(Numerical Answer Type)

This section contains **EIGHT** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. xxxxx.xx).

29. $NaOH \xrightarrow[Cold]{Cl_2} P + NaOCl + H_2O$

$NaOH \xrightarrow[Hot]{Cl_2} P + NaClO_3 + H_2O$

What is the molar mass of 'P' in gram unit?

30. In how many kg water, should 50 moles of $CaCl_2$ be dissolved in order to lower its freezing point by 5^oC? [Assume complete dissociation of $CaCl_2$]
[K_f of water = 1.86 K kg mol^{-1}]

Space for Rough work

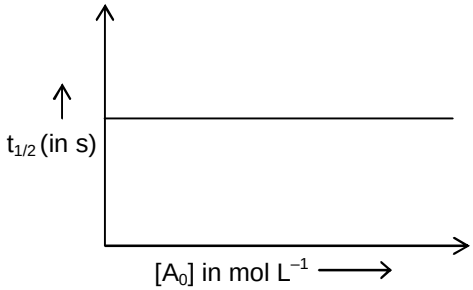
31. Percentage yield of 2-chlorobutane on photochlorination of n-butane is

32. The polymerization of the ethylene to linear polyethylene is represented by the reaction

$$n\left(CH_2 = CH_2\right) \longrightarrow \left(CH_2 - CH_2\right)_n$$

Where n has a large integral value. Given that the average enthalpies of bond dissociation for C = C and C – C at 298 K are +590 kJ/mol & +331 kJ/mol respectively. Calculate the enthalpy of polymerization in kJ per mole of ethylene at 298 K?

33. For a certain chemical reaction, the variation of half-life($t_{½}$) with different initial concentration(A_0) of reactant has been represented in the following plot.



What will be the half-life period of the reaction in 's' unit if the rate constant is $15 \times 10^{-3}\ s^{-1}$?

34. An orange colour solid(X) upon heating produces a green residue(Y) and evolves a colourless and odourless gas(Z). When treated with aqueous NaOH, (X) produces NH_3 gas and the solution turns yellow. What is the molar mass of 'X' in gram unit if the molar mass of 'Y' is 152 g?

35.

$$CH_2 = CH - CH_2 - \overset{\overset{O}{\|}}{C} - CH_2 - CH = CH_2$$

Above compound undergoes normal acidified permanganate(MnO_4/H^+) oxidation to form compound(X), CO_2 and H_2O. (X) is a dicarboxylic acid which on heating forms(Y). Reaction of three moles of (Y) with conc. H_2SO_4 forms compound(Z). What is the molar mass of (Z) in gram unit?

36. 4 g of nitrate of a univalent metal(M) was heated with excess of previously ignited silica. A loss of mass of 1.08 g takes place due to total expulsion of the nitrate part as N_2O_5. If the mass percentage of nitrate group in the salt is X%, what is the value of 'X'?

$$2MNO_3 + SiO_2 \longrightarrow M_2SiO_3 + N_2O_5$$

Space for Rough work

Mathematics

PART – III

SECTION – A
(One or More than one correct type)

This section contains **6** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

37. Tangent drawn at point P(1, 3) of a parabola intersects its tangent at vertex at M(–1, 5) and cuts the axis of parabola at T. If R(–5, 5) is a point on SP; where S is focus of the parabola, then
(A) slope of axis is –3
(B) radius of circumcircle of ΔSMP is $\sqrt{\frac{5}{2}}$ units
(C) $(ST)^2 - (SM)^2 = (PM)^2$
(D) tangent cuts the axis of parabola at T (–3, 7)

38. If y = f(x) and y = g(x) are symmetrical about the line x = $\frac{\alpha+\beta}{2}$, then $\int_{\alpha}^{\beta} f(x)\, g'(x)\, dx$ is equal to
(A) $\int_{\alpha}^{\beta} f'(x)\, g(x)\, dx$
(B) $-\int_{\alpha}^{\beta} f'(x)\, g(x)\, dx$
(C) $\frac{1}{2}\int_{\alpha}^{\beta} (f(x)\, g'(x) - f'(x)\, g(x))\, dx$
(D) $\frac{1}{2}\int_{\alpha}^{\beta} (f(x)\, g'(x) + f'(x)\, g(x))\, dx$

39. Suppose that f(x) is a differentiable invertible function, $f(x) \neq 0$ and $h(x) = \int_1^x f(t)dt$. Given that $f(1) = f'(1) = 1$ and g(x) is inverse of f(x).,
Let $G(x) = x^2 g(x) - xh(g(x))$ $\forall x \in R$. Which of the following are correct:
(A) G'(1) = 2
(B) G'(1) = 3
(C) G"(1) = 2
(D) G"(1) = 3

Space for Rough work

40. If the probability of choosing an integer 'k' out of 2m integers 1, 2, 3,……..2m is inversely proportional to $k^4\,(1 \le k \le m)$. If x_1 is the probability that choosen number is odd and x_2 is the probability that chosen number is even, then

(A) $x_1 < \frac{2}{3}$ (B) $x_1 > \frac{1}{2}$

(C) $x_1 > \frac{2}{3}$ (D) $x_2 < \frac{1}{2}$

41. Which of the following functions are twice differentiable at x = 0?

(A) $f(x) = x\,|x|$ (B) $g(x) = [x^2]\tan^{-1}x - \{x^2\}\cot^{-1}x - [x^2]\frac{\pi}{2}$

(C) $h(x) = |\sin^2 x|$ (D) $h(x) = \begin{cases} x^4\cos\left(\frac{1}{x}\right); & x \neq 0 \\ 0 \quad\quad\quad\;\; ; & x = 0 \end{cases}$

42. Let $S_n = \sum_{r=1}^{n} \frac{r^3 + n^3}{n^4}$ and $T_n = \sum_{r=0}^{n-1} \frac{r^3 + n^3}{n^4}$ for $n = 1, 2, 3, ...$ Then

(A) $S_n > \frac{5}{4}$ (B) $S_n < \frac{5}{4}$

(C) $T_n > \frac{5}{4}$ (D) $T_n < \frac{5}{4}$

Space for Rough work

(Matching List Type)

This section contains **FOUR** questions. Each question has **TWO** matching lists: **LIST - I** and **LIST - II**. **FOUR** options are given representing matching of elements from **LIST - I** and **LIST - II**. **ONLY ONE** of these four options corresponds to a correct matching.

43. Match the list:

List - I		List - II	
(P)	Radius of the largest circle which passes through the focus of the parabola $y^2 = 4x$ and is completely contained in it, is	(1)	16
(Q)	If the shortest distance between the curves $y^2 = 4x$ and $y^2 = 2x - 6$ is d then d^2 is	(2)	5
(R)	Let AB be a focal chord of $y^2 = 12x$ with focus S. The harmonic mean of lengths of segments AS and BS is	(3)	6
(S)	Tangents drawn from P meet the parabola $y^2 = 16x$ at A and B. If these two tangents are perpendicular then the least value of $\sqrt{AB}$ is	(4)	4
		(5)	–1
		(6)	0

The correct option is

(A) P → 4; Q → 2; R → 3; S → 5 (B) P → 4; Q → 2; R → 3; S → 4

(C) P → 4; Q → 2; R → 1; S → 6 (D) P → 4; Q → 3; R → 6; S → 5

44. The angles of a triangle are in the ratio 2 : 3 : 7

List – I		List – II	
(P)	Smallest side = 2	(1)	Largest side $= \sqrt{3}+1$
(Q)	Smallest side $= \sqrt{2}$	(2)	Largest side $= \sqrt{6}+\sqrt{2}$
(R)	Semi-perimeter $= 3+\sqrt{3}+\sqrt{2}$	(3)	Largest side = 1
(S)	Area of triangle $= (\sqrt{3}-1)/4$	(4)	Largest side $= 2+2\sqrt{3}$
		(5)	No such triangle is possible
		(6)	Angles are in AP

The correct option is

(A) P → 2; Q → 1; R → 4; S → 3 (B) P → 2; Q → 1; R → 3; S → 4

(C) P → 1; Q → 2; R → 4; S → 3 (D) P → 2; Q → 1; R → 5; S → 6

Space for Rough work

45. Each of 4 boys attends a school gathering with both of his parents. The number of ways these 12 people can be divided into groups of three each group containing a boy, a male parents, a female parent such that

List – I		List – II	
(P)	No boy is with both of his parents in his group is N then N is	(1)	5
(Q)	Exactly two boy is with both their parents is N then N is	(2)	453
(R)	Two particulars boys is with both of his parents is N then N is	(3)	3
(S)	Every boy is with both his parents is N then N is	(4)	18
		(5)	1
		(6)	283

The correct option is

(A) P → 6; Q → 4; R → 3; S → 5 (B) P → 2; Q → 1; R → 1; S → 5

(C) P → 6; Q → 2; R → 1; S → 5 (D) P → 2; Q → 4; R → 3; S → 5

46. A (–2, 0) and B (2, 0) are two fixed points and P is a point such that PA – PB = 2. Let S be the circle $x^2 + y^2 = r^2$. Then match the following.

List-I		List-II	
(P)	If $r = 2$, then the number of points P satisfying $PA - PB = 2$ and lying on $x^2 + y^2 = r^2$ is	(1)	2
(Q)	If $r = 1$, then the number of points satisfying PA – PB = 2 and lying on $x^2 + y^2 = r^2$ is	(2)	4
(R)	For $r = 2$, then the number of common tangents is	(3)	0
(S)	For $r = \frac{1}{2}$, then the number of common tangents is	(4)	1
		(5)	5
		(6)	6

The correct option is

(A) P → 6; Q → 4; R → 3; S → 5

(B) P → 2; Q → 1; R → 1; S → 5

(C) P → 1; Q → 4; R → 3; S → 1

(D) P → 2; Q → 4; R → 3; S → 5

Space for Rough work

SECTION – D
(Numerical Answer Type)

This section contains **EIGHT** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. xxxxx.xx).

47. Given the equation of the ellipse $\frac{(x-3)^2}{16}+\frac{(y-4)^2}{49}$ = 1, a parabola is such that its vertex is the lowest point of the ellipse and it passes through the ends of the minor axis of the ellipse. The equation of the parabola is in the form $16y = A(x-H)^2 - K$. Determine the value of $\frac{A}{7}+\frac{H}{3}+\frac{K}{16}$ is equal to

48. A straight line cuts the x-axis at point $A(1,0)$, and y-axis at point B, such that $\angle OAB = \alpha\left(\alpha > \frac{\pi}{4}\right)$. C is middle point of AB and B' is a mirror image of point B with respect to line OC and C' is a mirror image of point C with respect to line BB', then the ratio of the areas of triangles ABB' and BB'C' is equal to

49. The number of functions from $f : A \to B$ where $A = \{a_1, a_2, a_3, a_4, a_5, a_6\}$, $B = \{1, 2, 3, \ldots, 9\}$ such that $f(a_{i+1}) > f(a_i)\ \forall\ i \in N$ and $f(a_i) \neq i$ is

50. Let $(x^3 + px^2 + 2x - 5)^{19}(x^2 + qx - 41)^8(x^4 - x^3 + x - 7)^6 = x^{97} + 391x^{96} + a_{95}x^{95} + a_{94}x^{94} + \ldots. a_1x + a_0$ be an identity, where p, q, $a_{95}....a_0$ are integers. Compute the smallest positive value of p.

Space for Rough work

51. If number of points common between ||x| - |y|| = 1 and 2a |x| |y| + 1 = 2 |x| + a|y| are 8 then value of 3a (where a < 1) is ____.

52. Let f(x) = 30 – 2x – x^3, then find the number of positive integral values of 'x' which satisfies f(f(f(x))) > f(f(-x)).

53. $\vec{a}, \vec{b}$ and $\vec{c}$ be three non – coplanar vectors, and $\vec{d}$ be a non zero vector, which is perpendicular to $\left(\vec{a}+\vec{b}+\vec{c}\right)$. Now if $\vec{d}=\sin x\left(\vec{a}\times\vec{b}\right)+\cos y\left(\vec{b}\times\vec{c}\right)+2\left(\vec{c}\times\vec{a}\right)$, then minimum value of x^2+y^2 is equal to $\frac{\lambda\pi^2}{4}$, where λ is

54. If $\int \frac{(x-1)dx}{(x+x\sqrt{x}+\sqrt{x})\sqrt{\sqrt{x}(x+1)}} = 4\tan^{-1}[g(x)]+C$, where C is an arbitrary constant of integration. Find g^2 (1).

Space for Rough work